PRACTICAL-10

1.Aim

Design loT based home automation system using Raspberry Pi.

2. Prior Concepts

i) Concept of Python Programming

ii) Working of relay modules

iii) Basic networking concepts

3. New Concepts

i) loT-based remote appliance control

ii)Interfacing relay modules with Raspberry Pi

iii) Web-based appliance automation

iv) Flask web server programming

v)GPIO control using Python

vi) Smart home automation concepts

4. Hardware Required

i) Relay Module (2-channel)

ii)Blub

iii)DC Fan

iv) Jumper Wires

5.Procedure

Pin Connection

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

pip install flask

pip install RPi.GPIO

6. Implementation with Python Code

Code:

```
from flask import Flask
import RPi.GPIO as GPIO
```

```
app = Flask(name)

#GPIO Setup
GPIO.setmode(GPIO.BCM)

LIGHT = 17
FAN = 27
GPIO.setup(LIGHT, GPIO.OUT)
GPIO.setup (FAN, GPIO.OUT)

GPIO.output(LIGHT, GPIO.LOW)
GPIO.output(FAN, GPIO.LOW)

@app.route('/')
def home():
        return "'
        <h1>loT Home Automation</h1>

        <h3>Light </h3>
        <a href="/light/on">ON</a><br><br>
        <a href="/light/off">OFF</a><br><br>
        <h3>Fan</h3>
        <a href="/fan/on">ON</a><br><br>
        <a href="/fan/off">OFF</a>
        ‘‘
@app.route("/light/on')
def light_on():
GPIO.output(LIGHT, GPIO.HIGH)
return "Light ON"
```

```
@app.route('/light/off')
def light_off():
        GPIO.output(LIGHT, GPIO.LOW)
        return "Light OFF"

@app.route('/fan/on')
def fan_on():
        GPIO.output(FAN, GPIO.HIGH)
        return "Fan ON"

@app.route('/fan/off')
def fan_off():
        GPIO.output (FAN, GPIO.LOW)
        return "Fan OFF"

if name ==' _main':
app.run(host='0.0.0.0', port=5000)
```

7. Execute the above Program:

i)Run the above Program

ii) Find Raspi IP Address, by executing below command on command prompt:

hostname -I

Eg: 192.168.1.10

iii) Open Web Interface on mobile or laptop and enter the IP address in browser

Eg: http://192.168.1.10:5000

8. Result: